

Toronto Fresh Food (TFF)

**Local LLM AI Agent - README**

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## 1. Choosing the Best Local LLM Model to Download

Before choosing a local LLM runner, we have to first choose a model that we have to download. The following section goes over which local LLM model is best for the company's needs.

To choose the best local LLM model for the company, we need to look at the company's needs and restrictions. We ideally want a local LLM model that is best suited for agentic tasks and is able to create apps with great user interfaces (UI) and user experiences (UX) through powerful UI/UX generation. We are restricted to 32 GB of video random access memory (VRAM), meaning that powerful local LLM models like Kimi-K2.6 will not be possible to run on our devices because they require 250+ GB of VRAM. Based on the needs above, the best local LLM model to choose would be [Qwen3.5-27B](#) or Qwen3-Coder 30B (A3B) since they excel at handling multi-file refractors, understanding complex project structures and producing high quality UI code while also being good at agentic tasks. It would make the most sense to run these local LLM models on Q4\_K\_XL or 5/6 quants because they would be able to best compress the model so that it can run smoothly on just 32 GB of VRAM.

## 2. Choosing the Best Local LLM Runner

Now that we have chosen the best Local LLM model to use, we now need to choose the best local LLM runner for these models. I recommend [LM Studio](#) because it is the most user-friendly interface, allows for you to search for and download quantized GPT-generated unified formatted (GGUF) models directly, and tells you whether or not a local LLM fits within the memory on your device. The following [YouTube video](#) showcases how to run a local LLM model on LM Studio.

### How do we run Qwen3.5-27B in LM Studio?

#### 1. Download the model

1. Open LM Studio and use the search bar to search for:

unsloth Qwen3.5-27B-GGUF

2. Select the repository (usually by Unsloth) and locate the Q4\_K\_XL or UD-Q4\_X\_L file variant
3. Download the GGUF File

## 2. Configure system settings

1. GPU offloading: open the settings panel on the right, enable GPU offload, and set it to the max (99-100) to maximize the GPUs VRAM
2. Context Length: set the desired context length. The model supports up to 256K, but you should adjust this based on your available memory
3. KV Cache: if memory issues arise, consider using q8\_0 for your K and V cache, or reduce the context length

## 3. Run the model: load the model in the chat tab

### 3. Orchestration (Tools Connection)

If we were to use the following setup of running Qwen3.5-27B on Q4\_K\_XL quants on LM Studio, we could orchestrate the local LLM using the following process:

#### 1. Enable the LM Studio Server

- Navigate to the Developer/Local Server tab (represented by the <-> icon)
- Click “Start Server”
- Note your base URL. The default URL is usually `http://localhost:1234/v1`

#### 2. Choose Your Orchestration Framework

- Since our local server mimics OpenAI, we can use industry-standard orchestration tools. Since we want Retrieval-Augmented Generation (RAG) pipelines, using [LangChain](#) would be the most efficient option

#### 3. Connect the Orchestrator via Code

- I recommend using [Python](#) due to its simplicity and its compatibility because it is the most commonly used programming language in data science and data engineering
- You need to install Python SDK and LangChain with its Partner Packages using your command prompt/terminal by typing the following commands

(note: you will need to have Python installed on your machine for this to work):

```
pip install lmstudio
```

```
pip install langchain
```

```
pip install langchain-openai
```

```
pip install langchain-community
```

```
pip install langchain-chroma
```

The vanilla Python script you should run to connect LangChain to LM Studio is:

```
from langchain_openai import ChatOpenAI
from langchain_core.prompts import ChatPromptTemplate

# Point LangChain to LM Studio
llm = ChatOpenAI(
    base_url="http://localhost:1234/v1",
    api_key="lm-studio",
    model="qwen3.5-27b",
    temperature=0.7
)

prompt = ChatPromptTemplate.from_messages([
    ("system", "You are a helpful orchestration assistant."),
    ("user", "{input}")
])

chain = prompt | llm
print(chain.invoke({"input": "Explain how you will manage my tasks."}))
```

- Because LM Studio's local servers mimics OpenAI's API, you can just simply use LangChain's OpenAI classes and point them to your localhost
- This code is used to build tool-calling chains or RAG

## 4. Vector Database (Storage)

Using the same system above, here is how you can build a vector database.

### 1. Download an embedding model in LM Studio

- Open LM Studio and go to the search tab
- Choose an highly-rated embedding model, I recommend [nomie-embed-text-v1.5](#)
- Download a GGUF file for the embedding model
- Go to the "Local Server" tab in LM Studio. Ensure that you load both the Qwen model and your embedding model (LM studio allows for multiple models to be loaded within the server settings)

### 2. Set Up Your LangChain Environment

- You will need to install a vector store library on your device. **ChromaDB** is one of the easiest ones to use with Python. To download it, open your command prompt/terminal and install it using the following command:

```
pip install chromadb langchain-openai  
langchain-community sentence-transformers
```

- You will also need to install HuggingFaces' sentence transformers in the same manner:

```
pip install langchain-huggingface  
sentence-transformers
```

### 3. Write the Python Code to Build the Database

The vanilla Python script you should run to build the vector database is:

- We will use metadata filtering to ensure that all 8 of the departments: marketing, business development, production, accounting, HR, QA, engineering and warehousing will only receive answers that are retrieved from their departments documents in separate RAG pipelines

```
from langchain_huggingface import
HuggingFaceEmbeddings

# This downloads a small, fast, and popular
open-source embedding model
embeddings =
HuggingFaceEmbeddings(model_name="all-MiniLM
-L6-v2")

from langchain_openai import ChatOpenAI
from langchain_community.vectorstores import
Chroma

# 1. Initialize your LLM and Vector DB
LM_STUDIO_URL = "http://localhost:1234/v1"
llm = ChatOpenAI(
    openai_api_base=LM_STUDIO_URL,
    openai_api_key="lm-studio",
    temperature=0.3
)

# Assuming 'embeddings' is already defined
in your code
vector_db =
Chroma(persist_directory="./my_local_vectord
b", embedding_function=embeddings)

# 2. Define your list of departments exactly
as they appear in your metadata
departments = [
    "marketing",
    "business development",
    "production",
    "accounting",
    "HR",
    "QA",
    "engineering",
    "warehousing"
]
```

```

# 3. Create a dictionary to store a
dedicated retriever for each department
department_retrievers = {}

for dept in departments:
    dept_filter = {"department": dept}
    department_retrievers[dept] =
vector_db.as_retriever(search_kwargs={"k":
3, "filter": dept_filter})

# 4. Example Usage:
# If you want to search only within the HR
department's documents:
# hr_retriever = department_retrievers["HR"]
# hr_docs = hr_retriever.invoke("What is the
standard onboarding process?")
# If you want to search only within
Engineering:
# engineering_retriever =
department_retrievers["engineering"]
# eng_docs =
engineering_retriever.invoke("Where are the
deployment logs stored?")

```

## 5. Embedded Model (Data to a Vector)

Using the same system as before, here is how we can take the data and turn it into a vector.

1. Install the following package the same way you installed ChromaDB when making the vector database:

```

pip install langchain langchain-openai langchain-community
chromadb tiktoken

```

- In order for all types of documents to work, we need to install a few more libraries using command prompt/terminal again:

```

pip install unstructured pdf2image pdfminer.six
pip install pi-heif
pip install unstructured-inference

```

2. Write and run the following Python script to ingest your data into ChromaDB:

- Note: this is a skeletal program that will have to be changed and run each time we want to ingest a file
- “DepartmentName” should be replaced with what department you are in. Make sure that you write the department name in all lowercase letters except for HR and QA
- “FolderName” should be replaced with what folder you have the files stored in and it must be in the same folder as what the EmbeddedModel.py code is saved in

```
from langchain_community.document_loaders import
DirectoryLoader
from langchain_text_splitters import
RecursiveCharacterTextSplitter
from langchain_openai import OpenAIEmbeddings
from langchain_community.vectorstores import
Chroma

# --- 1. Load the Data ---
# This will automatically scan the folder, detect
the file types,
# and load all supported files it finds!
loader = DirectoryLoader('./FolderName',
loader_kwargs={"strategy": "fast"})
documents = loader.load()

# ---> ADD YOUR METADATA HERE <---
# Loop through all the loaded documents from the
folder
# and add your custom key-value pair to the
metadata dictionary.
for doc in documents:
    doc.metadata["department"] = "DepartmentName"

print(f"Loaded {len(documents)} document(s) and
assigned metadata.")

# --- 2. Chunk the Data ---
```



```

text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,
    chunk_overlap=50,
    separators=["\n\n", "\n", " ", ""]
)
# The splitter will automatically attach
{"department": "DepartmentName"} to every
resulting chunk!
chunks = text_splitter.split_documents(documents)
print(f"Split data into {len(chunks)} chunks.")

# --- 3. Initialize the Embedding Model via LM
Studio ---
embeddings = OpenAIEmbeddings(
    openai_api_base="http://localhost:1234/v1",
    openai_api_key="lm-studio",
    model="nomic-embed-text-v1.5",
    check_embedding_ctx_length=False
)

# --- 4. Generate Vectors and Store in ChromaDB
---
vector_db = Chroma.from_documents(
    documents=chunks,
    embedding=embeddings,
    persist_directory="./local_chroma_db"
)

print("Data successfully embedded and saved to
ChromaDB with metadata!")

```

- Once you run this script, a new folder called `./local_chroma_db` will appear in the project's directory. This holds your embedded vectors. When you are ready to chat with Qwen3.5 again using LangChain, just load the existing ChromaDB instance as a "retriever" and pass the retrieved chunks to the Qwen model as context

## 6. User Interface (LLM Interface)

1. Choose a Python User Interface (UI) Framework for LLMs. I recommend **Streamlit** because it is the industry standard for building AI data and apps quickly. It also has built-in chat elements like `st.chat_message` and `st.chat_input`, that make creating an UI similar to ChatGPT's very easy. As a result, I will show the rest of this section using Streamlit.
2. Install Streamlit and LangChain Chroma the same way you installed ChromaDB:

```
pip install streamlit
pip install langchain-chroma
```

3. Create your UI script, `UIInterface.py`

```
import streamlit as st
from langchain_openai import ChatOpenAI,
OpenAIEmbeddings
from langchain_chroma import Chroma
from langchain_core.messages import HumanMessage,
AIMessage, SystemMessage

# --- 1. Page Configuration ---
st.set_page_config(page_title="TFF AI Agent",
page_icon="TFF Logo.jpg")
st.title("Toronto Fresh Food (TFF) AI Agent. Ask
me any questions about your department at TFF!")

# --- 2. Sidebar: Department Filter ---
st.sidebar.title("Search Settings")
st.sidebar.markdown("Select a department to
filter the documents the AI reads.")
departments = [
    "engineering",
    "HR",
    "marketing",
    "sales",
    "QA",
    "accounting",
    "business Development",
```

```

        "production",
        "warehousing"
    ]
    selected_dept = st.sidebar.selectbox("Department
Filter", departments)

# --- 3. Initialize Models & Database ---
@st.cache_resource
def get_llm():
    return ChatOpenAI(
        base_url="http://localhost:1234/v1",
        api_key="lm-studio",
        model="qwen3.5-27b",
        temperature=0.7
    )

@st.cache_resource
def get_vector_db():
    # Make sure this perfectly matches the
embedding setup we built for ingestion!
    embeddings = OpenAIEmbeddings(

openai_api_base="http://localhost:1234/v1",
        openai_api_key="lm-studio",
        model="nomic-embed-text-v1.5",
        check_embedding_ctx_length=False
    )
    # Point this to the folder where your
database was saved
    return
Chroma(persist_directory="./local_chroma_db",
embedding_function=embeddings)

llm = get_llm()
vector_db = get_vector_db()

# --- 4. Manage Chat History in Session State ---
if "messages" not in st.session_state:
    st.session_state.messages = []

# Display existing chat messages

```

```

for message in st.session_state.messages:
    role = "user" if isinstance(message,
HumanMessage) else "assistant"
    with st.chat_message(role):
        st.markdown(message.content)

# --- 5. Handle User Input & RAG Logic ---
if prompt := st.chat_input("Ask a question based
on your documents..."):

    # Add user message to UI and session state

st.session_state.messages.append(HumanMessage(con
tent=prompt))
    with st.chat_message("user"):
        st.markdown(prompt)

    # Generate response
    with st.chat_message("assistant"):
        response_placeholder = st.empty()

        with st.spinner(f"Searching
{selected_dept} documents..."):
            # A. Retrieve relevant document
chunks based on the user's prompt and selected
department

            retriever = vector_db.as_retriever(
                search_kwargs={
                    "k": 3,
                    "filter": {"department":
selected_dept}
                }
            )
            context_docs =
retriever.invoke(prompt)

            # Combine the retrieved text chunks
into a single string
            context_text =
"\n\n---\n\n".join([doc.page_content for doc in
context_docs])

```

```

        # B. Create a hidden System Message
        that feeds the context to the AI
        system_instruction = f"""You are a
        helpful AI assistant for the {selected_dept}
        department.

        Use the following retrieved context
        from the company database to answer the user's
        question.

        If the answer is not contained within
        the context, simply say that you do not know.

        Retrieved Context:
        {context_text}
        """

        # Combine the system instruction with
        the ongoing chat history
        full_messages =
        [SystemMessage(content=system_instruction)] +
        st.session_state.messages

        # C. Call the LLM with the context and
        history
        with st.spinner("Generating answer..."):
            response = llm.invoke(full_messages)

        # Display response

        response_placeholder.markdown(response.content)

        # Optional: Add a dropdown menu to
        let the user see exactly which documents the AI
        read

        if context_docs:
            with st.expander("View Source
            Documents"):
                for i, doc in
                enumerate(context_docs):
                    st.markdown(f"**Source
                    {i+1}:**")

```

```

st.write(doc.page_content)
                                st.caption(f"Metadata:
{doc.metadata}")
                                st.divider()

# Add AI message to session state to remember
it for the next turn

st.session_state.messages.append(AIMessage(content=
response.content))

```

- Make sure that you save this program in the same folder that you saved “TFF Logo.jpg” in

#### 4. Run your UI

- You cannot run this code normally, you must use command prompt/terminal. In your command prompt/terminal, run:

```

cd C:\Users\User\Downloads\Local LLM AI Agent\
python -m streamlit run UserInterface.py

```

- This will automatically open a browser window with your new Chat UI connecting to LM Studio

## 7. Cloud Solution vs Physical Solution

If we wanted to use the Cloud Solution instead of directly running it from a device, we would need to rent a GPU-enabled virtual machine (VM) and then install the reference engine onto it. You will first have to choose a provider of GPU focused-cloud services or cloud virtual machines. I would recommend [Runpod](#) because of its second building, wide selection of GPUs (e.g; A100, H100, RTX 4090/5090) and is easy to use for AI workloads. Then you launch a VM (e.g; Ubuntu) equipped with NVIDIA GPUs and finally install LM Studio via the terminal commands on the cloud instance.

If we were to use Runpod to run LM Studio, the process would look something like this:

## 1. Spin up a base pod

- Log into Runpod at go to deploy
- Select a base template with CUDA support (e.g; the official runpod/base template for Ubtunu + CUDA)
- Select one of the GPUs that has 48 GB of VRAM (we need a buffer if multiple departments are going to use the local LLM) . I recommend **RTX 6000 Ada** because it is considered one of the best single-card solutions for running local LLMs due to its Ada Lovelace architecture that allows for fast inference speeds and high token generation
- (Important) - click “Customize Deployment” under “HTTP Ports” and enter “1234” (this is the default port LM Studio’s API uses)
- Deploy the pod

## 2. Connect and Install LM Studio CLI

- Once our pod is running, click “Connect” and open the Web Terminal (or SSH into it if you prefer)
- Install the LM Studio daemon by running the following command:

```
curl -fsSL https://lmstudio.ai/install.sh | bash
```

- Start the background service daemon:

```
lms daemon up
```

## 3. Download and Run the Model

- Find the Hugging Face Repository for Qwen3.5-27B with Q4\_K\_XL quants (see the link in first section)
- Download the model using LM Studio CLI:

```
lms get unsloth/Qwen3.5-27B-GGUF:Q4_K_XL
```

- Start the OpenAI-compatible API server:

lms server start

#### 4. Connect Our API

- To connect the external apps/scripts we made in the previous steps, we won't use localhost: 1234. Instead, Runplot provides a proxy URL to route external traffic to your Pod's dashboard
- Click "Connect" and look for the HTTP service link for port 1234
- You will use that URL as your base URL appending /v1 to the end. It will look something like the following:

`https://<YOUR_POD_ID>-1234.proxy.runpod.net/v1`

In order to share these local LLMs across TFF, we would need to use the following process that differs slightly whether or not we use a physical solution or a cloud solution.

For a physical solution we would use a host device (i.e. a central desktop) which is running LM Studio. Then, we would use the following steps to share the local LLMs across the company.

1. Ensure that LM Studio is downloaded on both the host and client machine
2. Active LM Link
  - Open the sidebar and click on "LM Link" and follow the prompts to log in
  - This creates a secure link across the networks
3. On our host device, load our desired model in the "AI Chat" or in the "Local Server" tab
4. Client access: on your remote device, open the LM link and select "Add a remote machine." The hosted models will appear as local options, allowing cross-network usage

This is the easiest and most secure way to share the local LLMs across the company if we intended to use one host machine.



For a cloud solution, we would use Runpod and use the following process:

1. Enable global networking: ensure your pod has global networking enabled
2. Install Tailscale on the Runpod container

1. Access the Runpod container via JupyterLab

2. Open a new terminal window

3. Install Tailscale:

- Run the official one-line installation script:

```
curl -fsSL https://tailscale.com/install.sh | sh
```

4. Start Tailscale:

- Once installed, start the tailscale daemon. Because Runpod containers might not have `systemd` enabled by default, run this command to start in the background:

```
sudo tailscaled --state=tailscaled.state &
```

5. Authenticate and Connect

- Run the following command to connect your Runpod container to your Tailscale network:

```
sudo tailscale up
```

- This will output a URL (e.g; <https://tailscale.com>)
- Copy and paste the URL into a browser and log in with someone's Tailscale account to authenticate the container

6. Verify connection

- To confirm that it is connected and to get the Tailscale IP address of your pod, run the following script:

```
sudo tailscale ip -4
```

3. Install [Tailscale](#) on all client devices
4. Accesses the Runpod LM Studio server using its Tailscale IP address (100.x.x)

The reason why it is preferred to use Tailscale over using Runpod's public URL method is that Tailscale is a mesh VPN. Public URLs expose your local service to the entire internet which can increase the risk of data prompt injection attacks, unauthorized access and scanning. A mesh VPN like Tailscale offers a zero-trust approach, restricting access to only authenticated devices. This makes it a more secure format for deploying the local LLMs.

Generally, running a local LLM on the cloud compared to running a local LLM from a central desktop is more expensive for consistent, high-volume use but is cheaper for occasional, short-term use. This decision comes from whether or not a CAPEX (local hardware purchase) is more preferable than an OPEX (cloud renting) scenario. If you decide to use an OPEX solution, you will have to pay per hour as well, per token or both. While this is cheaper for lower-frequency usage over a short period of time, as usage and time increases renting a GPU becomes more expensive than buying a GPU.

As for the security of the company, it would be preferable to use the physical solution because none of the company's data will ever leave company/trusted devices. This is preferable to the OPEX solution in which you will have to give TFF's data to a VM provider and have to trust that its security is strong enough to keep your data protected, especially because hackers have an easier time and more incentives to hack into a cloud provider than into your personal computer.

However, while the cloud solution may be more expensive than the physical solution, it may be more convenient. This is because with a physical solution, you have to ensure that the desktop stays on for all working hours because if it turns off, none of the client machines will be able to use the AI agents. Also, the cloud solution is more flexible because it is not attached to any expensive hardware that needs to be replaced. If we wanted to scale up the business with more powerful AI agents we wouldn't need to buy any new equipment, we would just have to cancel renting the current GPU and start renting a GPU with more VRAM. However, with the cloud solution the IT team would likely have to do more work since they would have to install Tailscale on the Runpod container and ensure that Tailscale is installed on all client machines alongside LM Studio. This is more complex than using the physical solution because LM Link is already built into LM Studio

so all the IT team would have to do is make sure that LM Studio is installed on all devices that want to use the AI agents.

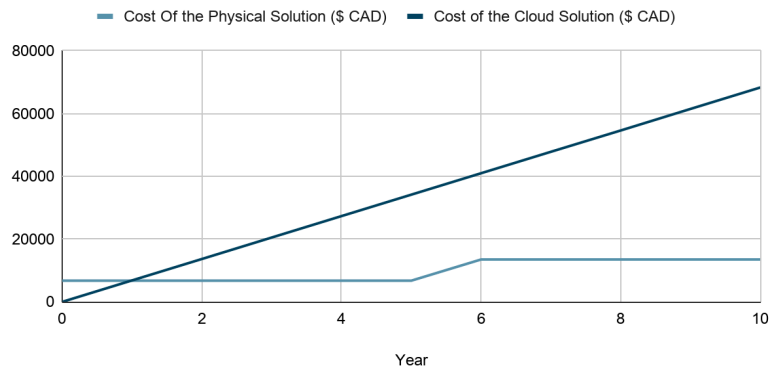
### What Would Be the Best Physical or Cloud Solution?

If we chose to use the physical solution it would make the most sense to buy a Mac desktop with 64-128 GB of RAM (once again, we need a buffer if all 7 departments are going to be using this Local LLM) and a monitor. The reason we should go with Mac over Windows in this case is because Apple Silicon uses unified memory between the CPU and the GPU, allowing for the GPU to utilize almost this entire pool of memory as VRAM. Also, we are only using these AI agents for casual, chat style interference. This means that we do not need the advantages that NVIDIA GPUs bring at training, high-context processing and diffusion image generation. I personally recommend the [Mac Studio with M3 Ultra Chip, 96 GB of Unified Memory and 1 TB of Storage](#) PC and [Alienware 34 Gaming - AW3425DWM](#) monitor. The total cost for this solution is \$6,756.26. The best cloud solution would be to use Runpod and rent the RTX 6000 Ada GPU and there would be no starting cost. However, it would cost \$0.78 per hour.

### Comparing the Best Solutions and Making a Final Decision

The following graph compares the total costs of the solutions over the span of ten years. The graph makes the assumption that the desktop and monitor are depreciating assets that have a useful life of 5 years and then need to be salvaged and replaced but salvage values were not deducted from the total costs. It also makes the assumption that we would be running the AI agents 24/7 on Runpod. Since we are using different RAG pipelines for each department, we do not need to deploy 7 Local LLMs, we only need to deploy one.

### Total Cost Of the Physical Solution and the Cloud Solution Over Time



As you can see from the graph, the physical solution over the span of ten years is cheaper at \$13,512.52 than the cloud solution at \$68,328. This means that on average, the cloud solution is about \$5,481.59 more expensive per year. As stated above, you can also salvage the PC and make some of your money back but you cannot do this with the cloud solution because you are renting a GPU and never really owned it as an asset. So, after examining the price differentials between both solutions in relation to TFF's revenue and profit margins over the next decade, the cloud solution is not meaningfully more expensive than the physical solution. Therefore, the price should not play a factor in the final decision.

As for what the final decision should be, it comes down to what the company prefers:

- If you prefer better security and easier to set up option at the expense of harder scalability and less flexibility, than the physical solution is best
- If you don't care as much about 100% security and are willing to spend more time setting up Tailscale on each client device for easier scalability and more flexibility, than the cloud solution is best

## 8. What if We Wanted to Use this Local LLM Model for General Inquiries?

Since Qwen 3.5-27B has already been trained, it will give answers based on its training to general inquiries in a process known as "hallucinating". The reason why it is not a good idea to use an AI model that hallucinates to give answers is

that they are not trained to the most relevant data. For example, the Qwen model we are using is only trained up till 2024, meaning that there will be much more relevant data that exists that the local LLM won't be able to use to give a precise answer. The whole purpose of using this model is to create a secure and stable way for each department at TFF to have an AI support agent that helps to answer their questions based on their department's documents. Trying to make a local LLM that can retrieve data from the internet is out of scope for the RAG pipelines we are using and users would be better off using a generative AI model from the internet such as Gemini Pro if they want to make general inquiries.

However, the AI agents can be good for automating some basic tasks based on their previous training. These tasks include: doing mathematical calculations, filling out forms and any other mundane tasks that do not require contemporary data to do. For example, if you want the sum of the costs of a bunch of sushi rolls, you can use the AI agent to calculate the sum for you if the software you are using does not have a calculator. Another relevant example could be that you want to improve the way an email/important document sounds, so you ask the Local LLM to improve your writing. This could be very useful because it allows you to get AI feedback on your writing without having to risk putting important company data on the internet if you were to use a standard LLM like Gemini Pro or ChatGPT.